Lab: Simple Linear Regression Analysis

Estimated time needed: 30 minutes

In this lab, you will learn how to perform a simple linear regression analysis using Microsoft Excel.

Software used in this lab

The instruction videos in this course use the full Excel Desktop version as this has all the available product features, but for the hands-on labs, we will be using the free "Excel for the web" version as this is available to everyone.

Although you can use the Excel Desktop software, it is recommended that you use Excel for the web for the hands-on labs, as the lab instructions specifically refer to this version, and there are some small differences in the interface and available features.

Data set used in this lab

The data set has been manually created.

Learning objective(s):

- Perform a simple linear regression analysis.

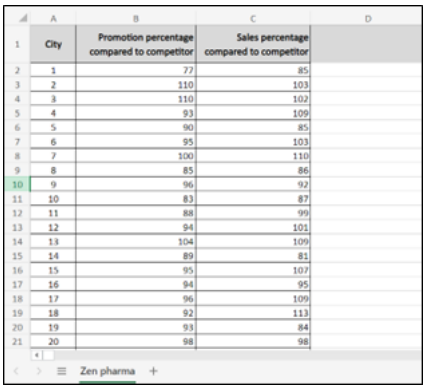
Welcome to the lab: Simple Linear Regression Analysis!

Exercise: Perform a simple linear regression analysis for sales versus promotion for Zen Pharma

Zen Pharma operates a chain of chemist shops across the country. The company has collected data from 50 randomly selected metropolitan cities to predict the impact of promotion on its sales. In each city, it has compared its promotional expenditures and sales with those of the leading competitors in the city over the past year.

Here's a screenshot of the data set showing the percentage of promotional expenditures incurred as compared to the leading competitor and the percentage of sales made compared to the competitor.

Download the file Zen Pharma_Promotions-Sales_Data.xlsx. To download the data set, right-click and open the link in a new tab. Open the file in Excel for the Web and follow the steps given below to perform the regression analysis.



City	Promotion percentage compared to competitor	Sales percentage compared to competitor
1	77	85
2	110	103
3	110	102
4	93	109
5	90	85
6	95	103
7	100	110
8	85	86
9	96	92
10	83	87
11	88	99
12	94	101
13	104	109
14	89	81
15	95	107
16	94	95
17	96	109
18	92	113
19	93	84
20	98	98

Zen pharma

Step 1: Identify the dependent and independent variables

We will predict the impact of the promotions on sales.

Therefore, the dependent variable is the sales percentage compared to the leading competitor, and the independent variable is the promotions percentage compared to the leading competitor.

As there is only one independent variable, the regression model will take the following form:

$Y = a + bX$

Y is the dependent variable, the sales percentage compared to the leading competitor
X is the independent variable, the promotions percentage compared to the leading competitor
a is the y-intercept
b is the slope-coefficient

Step 2: Determine the regression model

You can determine the regression model by finding a, the y-intercept, and b, the slope coefficient.

1. Calculate the y-intercept using the function **INTERCEPT(known_y's,known_x's)**, as shown below. This function applies only when there is a single independent variable.

G4 =INTERCEPT(C2:C51,B2:B51)

City	Promotion percentage compared to competitor	Sales percentage compared to competitor
1	77	85
2	110	103
3	110	102
4	93	109
5	90	85
6	95	103
7	100	110
8	85	86
9	96	92
10	83	87
11	88	99
12	94	101
13	104	109
14	89	81
15	95	107
16	94	95
17	96	109
18	92	113
19	93	84
20	98	98
21	103	112

Excel regression functions (When there is one independent variable only

Intercept: =INTERCEPT(C2:C51,B2:B51) = 25.13

2. Calculate the slope coefficient using the function **SLOPE(known_y's,known_x's)**, as shown below. This function applies only when there is a single independent variable.

G5 =SLOPE(C2:C51,B2:B51)

City	Promotion percentage compared to competitor	Sales percentage compared to competitor
1	77	85
2	110	103
3	110	102
4	93	109
5	90	85
6	95	103
7	100	110
8	85	86
9	96	92
10	83	87
11	88	99
12	94	101
13	104	109
14	89	81
15	95	107
16	94	95
17	96	109
18	92	113
19	93	84
20	98	98
21	103	112

Excel regression functions (When there is one independent variable only

Intercept: 25.126

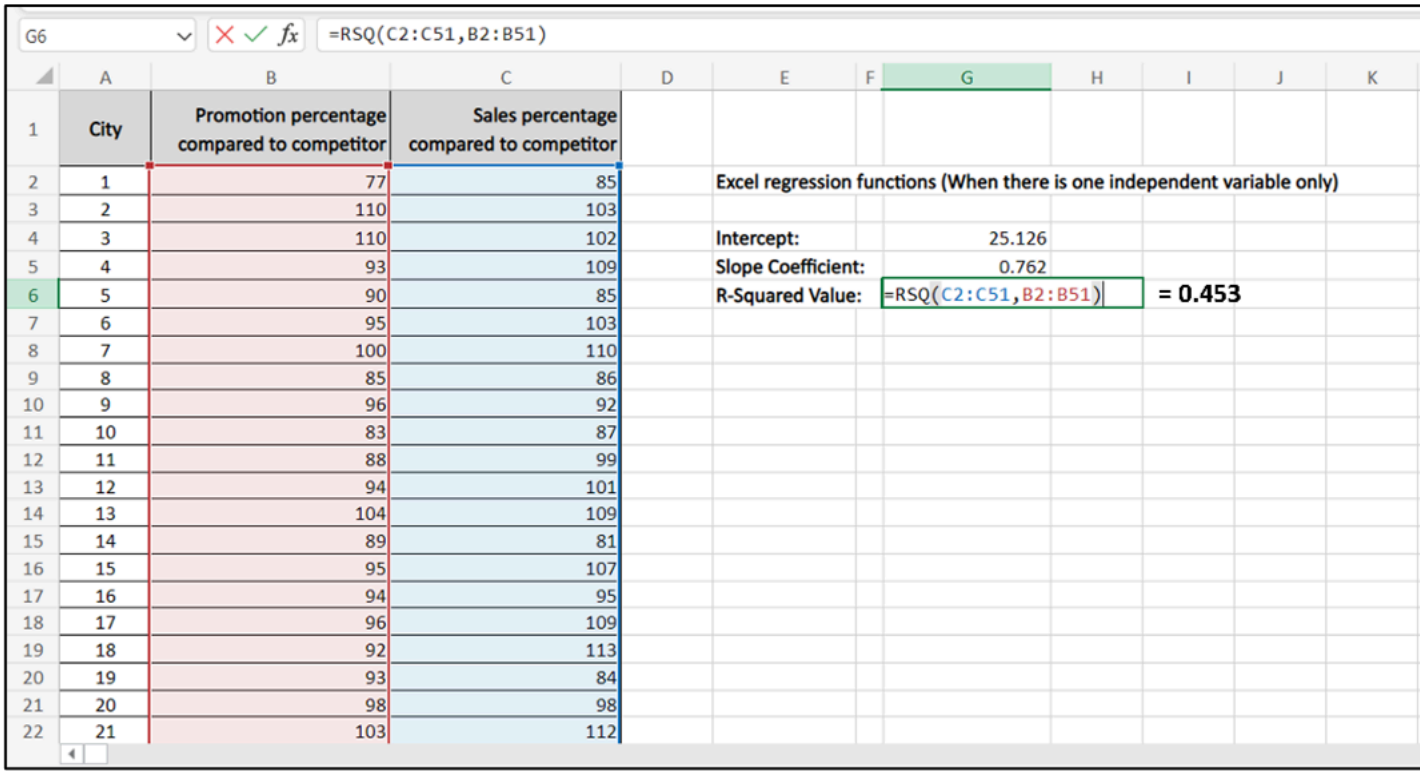
Slope Coefficient: =SLOPE(C2:C51,B2:B51) = 0.76

3. Therefore, the regression model can be represented as:

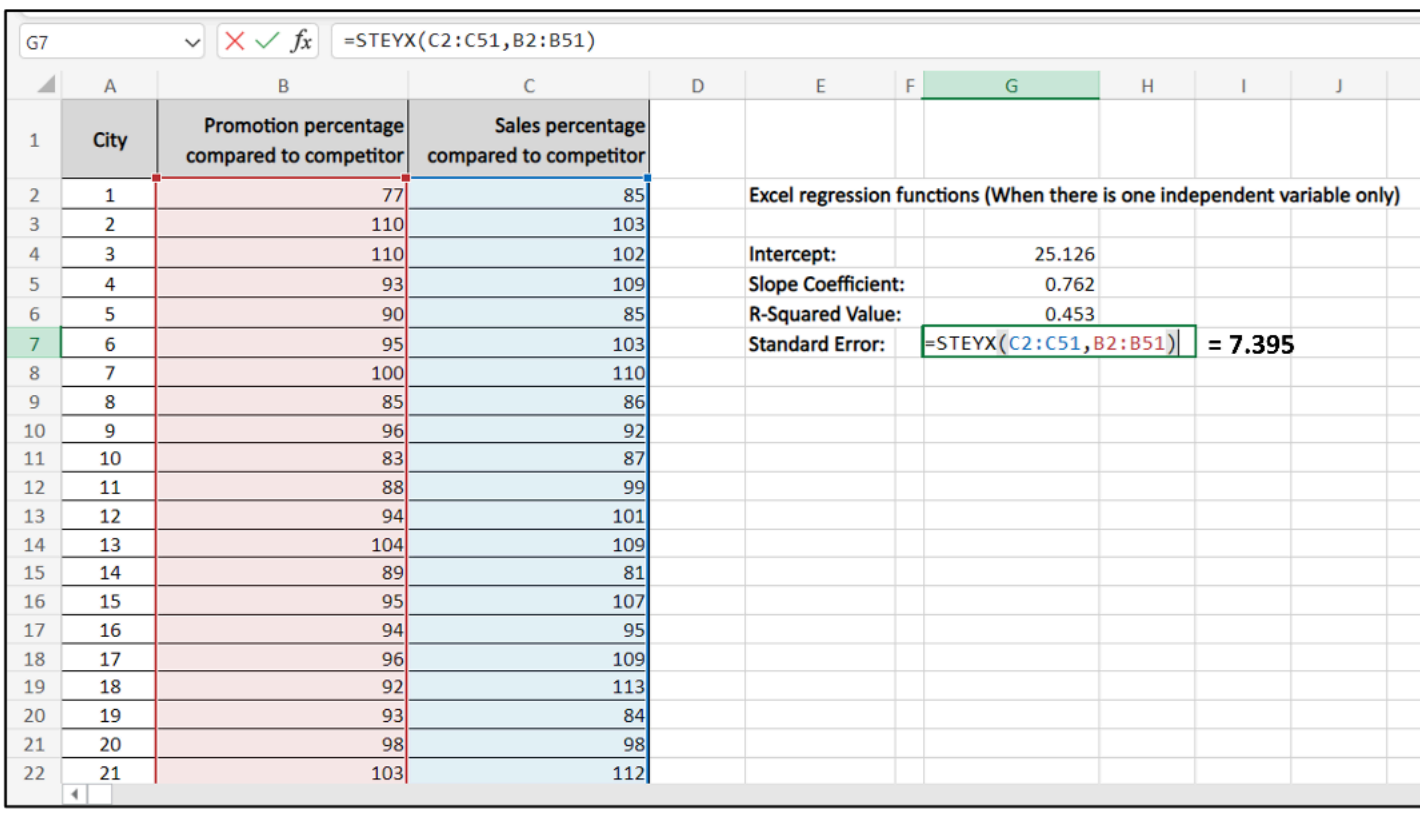
$Y = a + bX$
$Y = 25.126 + 0.762X$

Step 3: Evaluate the model

1. Find the R-squared value.
   To evaluate the model, find the R-squared value. To do so, use the function **RSQ(known_y's,known_x's)**, as shown below. The value 0.453 shows that 45.3% of the variance in the dependent variable is caused by the independent variable.



2. Calculate the standard error.
   To calculate the standard error, use the **STEYX(known_y's,known_x's)** function, as shown below. This value represents the accuracy of the model.

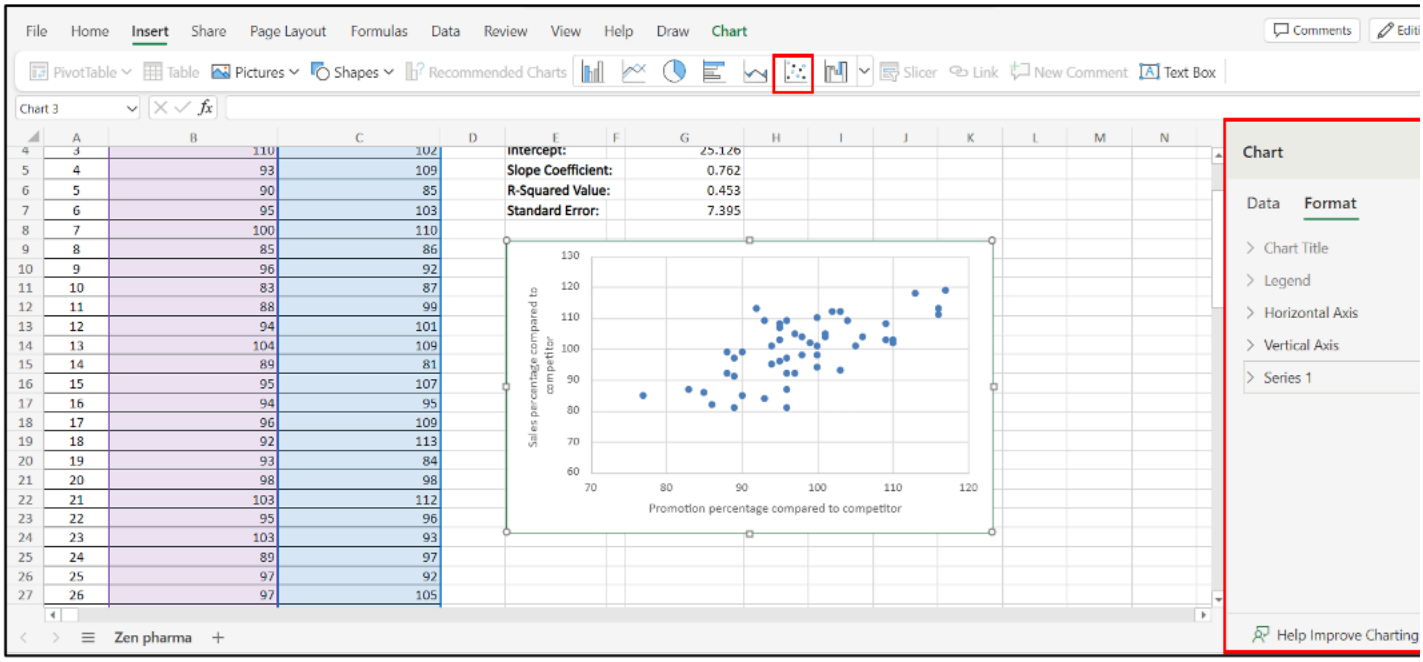


Alternative way of performing a simple linear regression and evaluating it

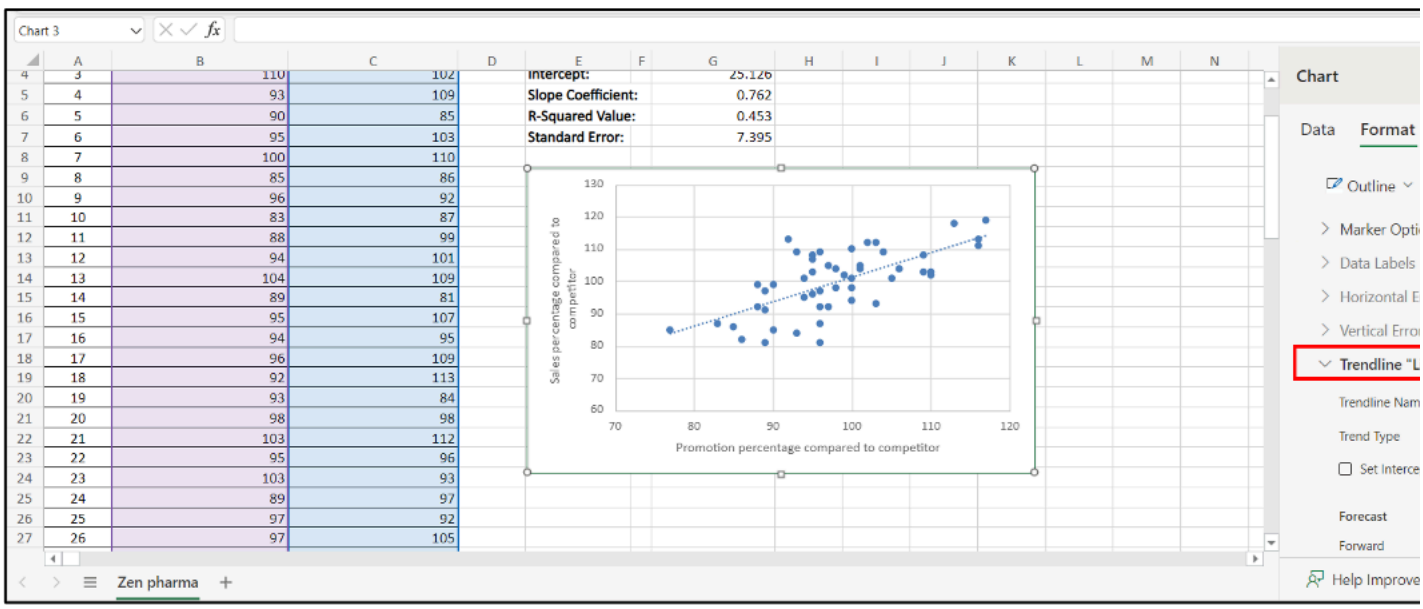
1. Insert a scatter chart for the given data. Plot the values in the "Promotion percentage compared to competitor" column along the x-axis, and plot the values in the "Sales percentage compared to competitor" along the y-axis.

   Use the settings panel to format the scatter plot as desired. In the image below, the x-axis and y-axis meet at the point (70, 60) instead of the default (0,0) to enable

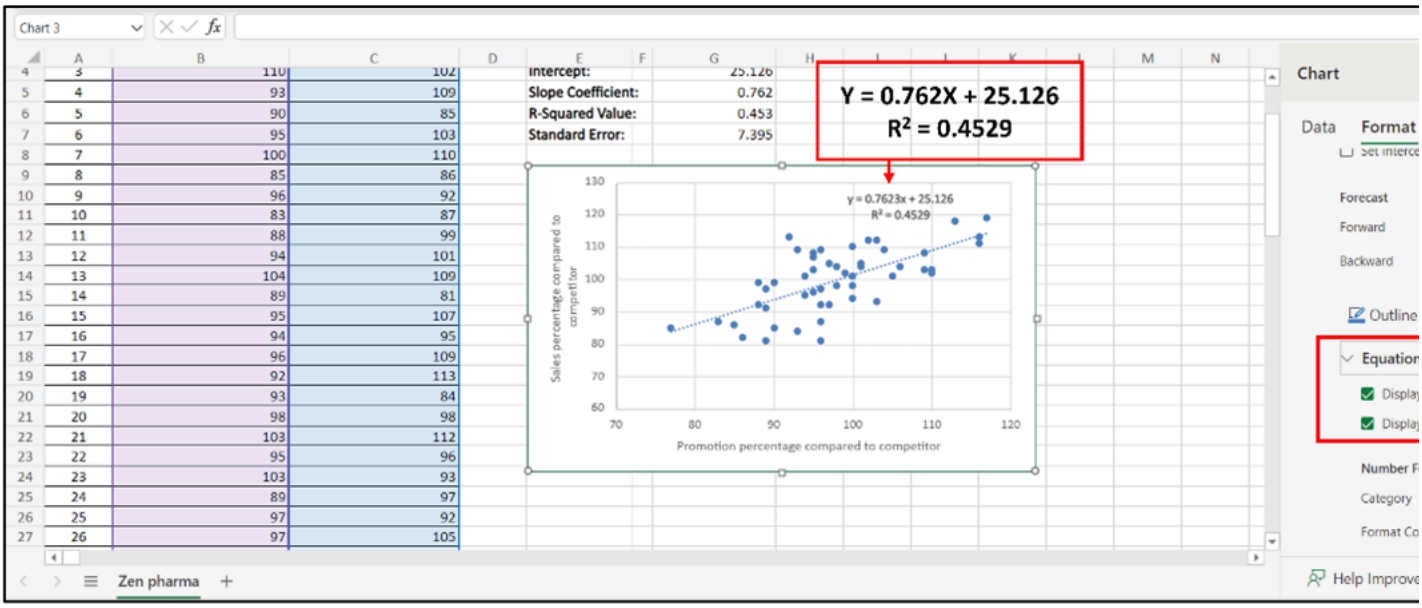
us to see the points clearly. This is done by setting the minimum bounds of the horizontal and vertical axes to 70 and 60, respectively. The titles of the axes have also been added.



2. Under the Series dropdown, activate the Trendline toggle to the "On" mode to display the trendline.



3. When you set the Trendline toggle to the "On" mode, the trendline settings will be displayed. Scroll down to Equation and Values and activate the toggle to "On." Select the **Display Equation on chart** and **Display R squared value on chart** checkboxes to display the equation and R-squared value on the chart.



Congratulations! You've completed the lab activity on simple linear regression, and you are ready to move to the next topic.

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